

12.CLINIC ON WHEELS AMBULANCE

TECHNICAL SPECIFICATION

1. BASE VEHICLE

- New, latest model& diesel vehicle
- Minimum **BS-VI** emission compliance.
- Minimum **2500 cc** engine or suitable equivalent.
- Power: Minimum **120 HP** or better.
- Manual/Automatic transmission.
- Power steering.
- ABS with EBD.
- Airbags for driver and co-driver.
- Central locking.
- Air conditioning for driver and patient compartment.
- Reverse parking camera and sensors.
- Tubeless tyres.
- High-intensity LED emergency beacon.
- Siren with public address system.
- GPS tracking system.
- Fire extinguisher.
- First-aid kit.
- Reflective markings and ambulance signage.

2. PATIENT COMPARTMENT

- Separate driver and patient compartments.
- Minimum **2.0 m internal patient compartment length** or as permitted by applicable vehicle regulations.
- Adequate headroom for attending medical staff.
- Anti-skid, seamless, washable flooring.
- Easy-to-clean, disinfectant-resistant interior panels.
- Medical-grade storage cabinets.
- Lockable drawers for medicines and consumables.
- Overhead storage compartments.
- IV fluid holders.
- Medical equipment mounting brackets.
- Adequate lighting with LED ceiling lights.
- Independent air-conditioning system for patient compartment.
- Exhaust/ventilation system.
- Grab rails and handrails.
- Privacy curtains.
- Seating for minimum 3 medical attendants.
- Foldable attendant seats with seat belts.

3. PATIENT STRETCHER

- Motorized or manual hydraulic stretcher.
- Adjustable backrest and leg section.
- Trendelenburg and reverse Trendelenburg positions.
- Minimum patient load capacity: **200 kg or better**.
- Locking mechanism for ambulance floor mounting.
- Safety straps.
- Side rails.
- Adjustable height.
- Mattress with waterproof, fire-retardant and antimicrobial cover.
- Compatible with CPR and shock positions.

4. DEFIBRILLATOR WITH PACING

- Biphasic defibrillator.
- Manual defibrillation.
- AED mode.
- Synchronized cardioversion.
- Transcutaneous pacing.
- Energy selection: Minimum **200 J or higher**.
- Adult and pediatric paddles/pads.
- ECG monitoring.
- Rechargeable battery.
- Built-in event recording.
- Minimum 3-lead ECG monitoring.
- SpO₂ and NIBP monitoring preferred.
- EtCO₂ monitoring preferred.

5. MULTIPARAMETER PATIENT MONITOR

Minimum parameters:

- 3/5-lead ECG.
- Heart Rate.
- SpO₂.
- NIBP.
- Respiratory Rate.
- EtCO₂.
- Temperature.
- Invasive BP (IBP) preferred.
- Minimum 12-inch color display.
- Trend storage.

- Audible and visual alarms.
- Rechargeable battery.

6. EMERGENCY VENTILATOR

- Microprocessor-controlled transport ventilator.
- Suitable for Adult and Pediatric patients.
- Volume Control and Pressure Control modes.
- SIMV/PSV.
- CPAP/PEEP.
- Tidal Volume: Approximately **50–2000 mL or better**.
- FiO₂: 21–100%.
- PEEP: Minimum 0–20 cmH₂O.
- Integrated oxygen supply.
- Internal rechargeable battery.
- High/low pressure alarms.
- Apnea alarm.
- Low battery alarm.
- Patient disconnect alarm.

7. OXYGEN SYSTEM

- Minimum **2 × 10 L medical oxygen cylinders** or equivalent capacity.
- Oxygen cylinders mounted securely.
- Pressure regulators.
- Flowmeters.
- Oxygen outlets.
- Oxygen tubing.
- Oxygen masks:
 - Adult
 - Pediatric
 - Neonatal
- Nasal cannulas.
- Non-rebreathing masks.
- Venturi masks.

8. SUCTION SYSTEM

- Electric medical suction unit.
- Vacuum range: Minimum **0–600 mmHg**.
- Flow rate: Minimum **30 L/min**.
- Collection bottle: Minimum 1 L.
- Disposable suction tubing.
- Yankauer suction tips.
- Portable battery-operated suction unit as backup.

9. AIRWAY MANAGEMENT EQUIPMENT

The ambulance shall include:

- Adult, Pediatric and Neonatal Ambu Bags.
- Oxygen reservoir bags.
- Face masks of multiple sizes.
- Laryngoscope with adult and pediatric blades.
- Video laryngoscope (preferred).
- Endotracheal tubes of different sizes.
- Cuffed and uncuffed tubes.
- Stylets.
- Bougie.
- Magill forceps.
- Supraglottic airway devices.
- LMA of multiple sizes.
- ET tube fixation devices.
- Oropharyngeal airways.
- Nasopharyngeal airways.
- Bite blocks.
- Capnography adapter.

10. INFUSION & DRUG DELIVERY

- Syringe Pump – Minimum 2 Nos.
- Infusion Pump – Minimum 2 Nos.
- IV stands.
- IV cannulas of various sizes.
- Central venous catheter kits.
- Pressure infusion bag.
- Three-way stopcocks.
- IV extension sets.
- Blood transfusion sets.

11. EMERGENCY & TRAUMA EQUIPMENT

- Spine board.
- Cervical collars – Adult and Pediatric.
- Scoop stretcher.
- Vacuum mattress.
- Splints – Adult and Pediatric.

- Pelvic binder.
- Tourniquets.
- Trauma dressing kits.
- Burn dressing kits.
- Hemorrhage control kit.
- Emergency blanket.
- Safety scissors.
- Tourniquet.

12. ECG MACHINE

- Portable **12-channel ECG machine**.
- Rechargeable battery.
- Thermal printer.
- Automatic ECG interpretation.
- ECG electrodes and cables.
- Minimum 100 ECG memory storage.

13. NEONATAL & PEDIATRIC EQUIPMENT

- Neonatal resuscitation bag.
- Pediatric Ambu Bag.
- Pediatric oxygen masks.
- Pediatric BP cuffs.
- Neonatal BP cuffs.
- Pediatric SpO₂ probe.
- Infant weighing scale.
- Neonatal transport incubator (optional/preferred for specialized ambulance).
- Neonatal suction equipment.

14. POINT-OF-CARE DIAGNOSTIC EQUIPMENT

- Glucometer.
- Portable blood gas analyzer .
- Thermometer.

15. MEDICAL STORAGE

- Lockable medicine cabinet.
- Controlled drug storage.
- Emergency drug tray.
- Sharps disposal container.

- Biohazard waste container.
- IV fluid storage.
- Consumable storage.
- Medication refrigerator (optional).

16. COMMUNICATION SYSTEM

- Mobile communication system.
- GPS tracking.
- Two-way communication system.
- Bluetooth/USB connectivity.
- Wi-Fi connectivity preferred.
- Emergency communication with hospital control room.

17. ELECTRICAL SYSTEM

- 12/24 V DC vehicle electrical system.
- Additional medical-grade inverter.
- Minimum **2 kVA pure sine wave inverter**.
- 230 V AC power outlets.
- Auxiliary battery system.
- Battery management system.
- Automatic changeover.
- Surge protection.
- Adequate electrical outlets for all medical equipment.
- Electrical safety isolation.

18. EXTERNAL FEATURES

- Emergency LED light bar.
- Side emergency lights.
- Rear warning lights.
- Electronic siren.
- Public address system.
- Reflective "AMBULANCE" signage.
- Red cross/emergency medical symbols as permitted.
- Rear step.
- Rear and side access doors.
- Automatic door lighting.

19. INFECTION CONTROL

- Seamless washable flooring.
- Antimicrobial surfaces preferred.
- Easy-to-clean wall and ceiling panels.
- Hand sanitizer dispenser.

- Medical waste segregation.
- Sharps container.
- UV sterilization system (optional).
- Hand-washing facility with water tank and waste tank (preferred).

20. SAFETY & REGULATORY REQUIREMENTS

The ambulance shall comply with:

- **AIS-125** – Code of Practice for Type C/D Ambulances, as applicable.
- Applicable **Central Motor Vehicle Rules (CMVR)**.
- Applicable **MoRTH** requirements.
- **BS-VI** vehicle emission norms.
- Relevant BIS/IEC standards for medical equipment.
- Medical equipment shall be CE/FDA approved or comply with applicable Indian regulatory requirements.
- Equipment shall be installed securely with certified mounting systems.

21. WARRANTY & SERVICE

- Vehicle warranty: Minimum **3 years or 1,00,000 km**, whichever is earlier.
- Medical equipment warranty: Minimum **3 years comprehensive**.
- Preventive maintenance: Minimum 4 visits/year.
- 24×7 breakdown support.
- Maximum response time: 24 hours.
- Spare parts availability: Minimum 10 years.
- Comprehensive training for drivers, paramedics, nurses and hospital biomedical engineers.
- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.